MILITARY SPECIFICATION

MOUNT, PERISCOPE: M118 AND M118E1

## 1. SCOPE

1.1 This specification covers instruments known as Mount, Periscope: M118 and Mount, Periscope: M118E1 (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of |
| MIL-P-14232 | Parts, Equipment and Tools for Army Materiel; Packaging and Packing of |
| MIL-I-45607 | Inspection Equipment, Supply and Maintenance of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for inspection by Attributes |
| MIL-STD-109 | Quality Assurance Terms and Definitions |

DRAWINGS

Frankford Arsenal

| | |
|---|---|
| F8619450 | Mount, Periscope: M118 |
| A8644448 | Push-Pull Gage |
| F10554660 | Holding Fixture |
| F10554661 | Set-Up Final Test Fixture for Mount, Periscope: M118 |
| F10554663 | Test Standard for Mount Periscope: M118 |



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| | |
|---|---|
| F11732148 | Gage, Interchangeability, Maximum |
| F11732149 | Gage, Interchangeability, Minimum |
| E11727490 | Mount, Periscope: M118E1 |

PACKAGING DATA SHEET

| | |
|---|---|
| PS8619450 | Packaging of Mount, Periscope: M118 |
| PS11727490 | Packaging of Mount, Periscope: M118E1 |

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS.

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing F8619450 (M118), or E11727490 (M118E1), and drawings pertaining thereto.

3.2 General sPecification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.2.1 Interchangeability. - The mount shall be capable of permitting insertion of the gages specified in 4.6.7. After insertion, the gages shall be capable of being firmly secured and subsequently removed.

3.3 Environmental.

3.3.1 Operating temperatures.- The mount shall meet the applicable requirements of 3.5.1 to 3.5.4.2 inclusive while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.3.2 Vibration.- The mount shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 ± 1/4 minutes. Subsequent to vibration, there shall be no evidence of physical failure and the mount shall meet the following requirements of this specification (see 3.5.4.2).

3.4 Orientation.- The locating surface of the mounting flange of the mount shall be secured in a horizontal plane and the headrest assemblies shall be in their uppermost vertical position. Gages referenced in 3.2.1 shall be utilized to determine compliance with the requirements of 3.5.1 and 3.5.2.



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3.5 Performance.

3.5.1 Release grip.- With the gages in 3.2.1 inserted into the mount, the release grip shall automatically latch. the gages in the fully inserted position. A direct pull of not more than 15 pounds on the release grip shall actuate the grip to its full disengaged position. Full disengagement of the grip shall allow removal of the gages from the mount without restrictions.

3.5.2 Holding features.- When Lever 8619482 is pushed to its fullest forward position and locked by releasing the plunger, the gages (see 3.2.1) shall be secured in the mount. Subsequent to the application of a tem pound pressure applied horizontally to the front and rear of each gage at a point 7.5 inches below the center of the plugs (Part No. 8298440), the gage shall be within .0005 inch of its original location when measured horizontally at a point 7.5 inches below the center of the plugs. This requirement shall be met following the release of each of the pressures applied.

3.5.3 Shield assembly.- With the plunger depressed (disengaged), maximum operation of the handle shall operate the shield through full operation (open for viewing and closed for protection). When the shield is fully open and when the shield is fully closed, the plunger shall engage and lock the handle when the plunger is released.

3.5.4 Headrest.

3.5.4.1 Headrest horizontal travel.- Each Knob B8619466 (M118) or Screw B10516599 (Ml18El) shall be loosened and each headrest assembly shall be adjustable through maximum movement permitted by the elongated slot in the assembly without restriction or force.

3.5.4.2 Retention. - The torsional load applied to each Screw 8619463 (see applicable Drawing F8619450 or E11727490) shall be such that the headrests shall not change position when subjected to the vibration specified in 3.3.2 and shall be operable at the temperature specified in 3.3.1.

4. QUALITY ASSurance PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three mounts plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior



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to Government acceptance or rejection by item or lot.

4.3.2.2 100 Percent inspection.- The following requirements in Table I and related tests shall be subjected to 100% inspection:

TABLE I. - CLASSIFICATION OF DEFECTS
100% INSPECTION

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 101. | Interchangeability | 3.2.1 | 4.6.7 |
| 102. | Release grip | 3.5.1 | 4.6.3 |
| 103. | Holding features | 3.5.2 | 4.6.4 |
| 104. | Shield assembly | 3.5.3 | 4.6.5 |
| 105. | Headrest horizontal travel | 3.5.4.1 | 4.6.6 |
| 106. | Retention | 3.5.4.2 | 4.6.6.1 |

MINOR: None

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One mount shall be selected at ranks as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURES |
|---|---|---|---|
| 301. | Fabrication | 3.1 | Applicable drawing-Visual |
| 302. | General specification | 3.2 | MIL-F-13926 - Visual |

4.4.2 Environmental.- Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the rerquirements and tests in Table III.



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TABLE III

| NO. | CHARACTERISTIC | REQUIREMENT | TEST PROCEDURE |
|---|---|---|---|
| 303. | Release grip | 3.3.1, 3.5.1 | 4.6.1.1, 4.6.3 |
| 304. | Holding features | 3.3.1, 3.5.2 | 4.6.1.1, 4.6.4 |
| 305. | Shield assembly | 3.3.1, 3.5.3 | 4.6.1.1, 4.6.5 |
| 306. | Headrest horizontal travel | 3.3.1, 3.5.4.1 | 4.6.1.1, 4.6.6 |
| 307. | Retention | 3.3.1, 3.5.4.2 | 4.6.1.1, 4.6.6.1 |
| 308. | Vibration | 3.3.2 | 4.6.1.2 |

**NOTE: The tests for Characteristics 303 through 307 inclusive shall be conducted at -40° and +150°F.**

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within ten percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.



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4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 operating temPeratures.- All moving or movable parts shall be inspected for operability and performance when exposed to the specified low and high temperatures of 3.3.1. The release grip, holding features, shield assembly and headrest shall be inspected in accordance with 4.6.3 to 4.6.6.1 inclusive.

4.6.1.2 Vibration. - The mount shall be positioned in a vibration test adapter secured to the vibration equipment. The mount shall be vibrated in a vertical plane in accordance with the conditions specified in 3.3.2. A visual inspection shall be made for physical damage or loose components and assemblies. The mount shall then be subjected to the following tests specified herein.

4.6.2 Orientation. - Special test equipment conforming to the requirements of Drawing F10554660 (Holding Fixture) shall be utilized to accept the mount, completely assembled, and secure it in the position required by 3.4. Set-up of the test equipment and installation of the mount under test shall be performed in accordance with the applicable set-up instructions contained on Drawing F10554661 (Set-Up Final Test Fixture for Mount Periscope: M118).

4.6.3 Release grip.- When the mount is oriented and positioned on the device as specified in 4.6.2, the gages in 4.6.7 shall be utilized for this test. Release the Locking Lever 8619482 and pull to full open position. It shall remain at this position for this test. Insert the maximum interchangeability gage into the bottom of the mount and push up until the release lever is activated and snaps into position. Caution should be used to make sure the gage is secured before the inspector releases his hold on it, to prevent injury. Do not lock Lever 8619482. Allow the release grip (safety latch) to hold the gage in the mount. To remove the gage an upward pressure shall be applied with the left hand to prevent dropping. A pull of not more than 15 pounds on the release grip, using standard measuring equipment, shall allow the gage to drop free and be removed from the mount as specified in 3.5.1. Perform the above test using the minimum interchangeability gage to determine compliance with 3.5.1.

4.6.4 Holding features.- A test standard conforming to the requirements of Drawing F10554663 shall be utilized in lieu of the referenced. gages (see 3.2.1 & 3.5.2). With the mount positioned on the holding fixture, the Test Standard 10554663 shall be inserted into the mount utilizing the procedure described in 4.6.3. Subsequent to insertion of the test standard, the Locking Lever 8619482 shall be released and then pushed to its forward-most position. This action will lock the test standard in the mount and the spring loaded plunger will secure the lever assembly in a locked position. A dial indicator graduated in .0001 inch increments shall



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be positioned so that it contacts the front or the rear button provided on the left side of the test standard, and then adjusted to read zero with respect to the indicating hand. Apply a slow steady pressure of ten pounds, by means of a push-pull gage conforming to Drawing A8644448, to the opposite side of the dial indicator contact point; and then upon releasing the pressure, the indicating hand of the dial indicator shall not exceed the tolerance specified in 3.5.2. Reverse the equipment and repeat the above procedure. Upon satisfactory completion of this portion of the test, repeat the entire above procedure at the front and rear buttons on the right side of the test standard.

4.6.5 Shield assembly.- A manual inspection of the shield assembly mechanism shall be made. Press the plunger to release the mechanism, if the shield is covering the aperture, an upward push to raise the shield for operation must be made. When the mechanism is pushed to the maximum point the plunger will lock the shield in place. To close, press plunger and pull to closed and locked position. There shall be a free and unrestricted movement as required by 3.5.3.

4.6.6 Headrest horizontal travel.- The manual testing of the headrest assemblies can be accomplished at the same time 4.6.6.1 is being performed. The assemblies shall meet the requirements of 3.5.4.1.

4.6.6.1 Retention. - The torsional load applied shall not restrict the headrest from being adjusted to the full up or down positions by hand and must stay positioned as specified in 3.5.4.2.

4.6.7 Interchangeability.- This test shall be performed utilizing the maximum and minimum gages conforming to Drawings F11732148 and F11732149 respectively, to determine compliance with 3.2.1.

## 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8619450 or 11727490. The level of protection shall be as specified in the procurement document.

## 6. NOTES

6.1 Intended use. - The Periscope Mount: M118 is used to support either the Periscope, Tank: M32 or Periscope, Tank: M35; the Periscope Mount: Ml18El supports either the Periscope, Tank: M32, or Periscope, Tank: M35E1.

6.2 Ordering data.- Procurement documents should specify the following:



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(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Provisions for First Article testing.

Custodian:

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Preparing activity:

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-45664E(MU) Mount, Periscope: M118 and M118E1

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO. |
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